Four-coordinate repair closure

1 November 2026

The field proof establishes that the literal four-coordinate matrix action preserves nonzero vectors when $M^2 = cI$ and $c \neq 0$. It also establishes double-action projective identity and preservation/reflection of the cross-product comparator for nonzero inputs.

Connection to native code

`walk-source.rs.txt:48-50` folds four products into each row, starting at zero. The field definition preserves the indexing and addition grouping. In `xadd_repaired` (lines 77–95), a successful repaired branch receives nonzero z from `xadd` and returns Mz without a second nonzero check. The theorem supplies that field postcondition:

$$M(Mz) = cz, \qquad c \neq 0,\ z \neq 0 \quad \Longrightarrow \quad Mz \neq 0.$$

The theorem does not imply branch success or correct geometric differential addition. Native word operations and the complete fold's canonicality are separate implementation obligations.

Corrected chart basis

The copied N16 note distinguishes the degree-two matrix C_2 from degree-one matrix C_1. The new exact field identity is

$$C_2(1, -a_1, -a_0, q_3)^T = C_1(1, -a_1, a_0, q_3 + \sigma a_0)^T,$$

assuming the recorded column relation $C_2[:, 2] = -C_1[:, 2] - \sigma C_1[:, 3]$ and equality of other columns. It does not justify using C_2 for degree one. A negative control shows that substitution can change the projective point.

Checks and evidence boundaries

Ten main declarations and nine rational controls are checked, including a doubled permutation with square $4I$, literal coordinates, two repairs, the $c = 0$ boundary, zero-vector comparator behavior and wrong-chart substitution. Controls use kernel tactics. Three failed construction attempts are retained. An early control build used an older module binary; the final ordered replay supersedes it and binds inputs before and after.

Historical N16 documents report 105 recorded scalar-square actions and chart certificates. No historical validator or raw-table audit was rerun. Concrete runtime action binding requires the exact modulus, matrix and nonzero scalar plus field/word correspondence. Geometric chart interpretation and xadd correctness remain separate premises.

`REPORT.md`, `status.json`, source hashes, native receipts and axiom readbacks provide the review record. This is a producer packet awaiting independent review and publication binding. No scientific execution, new measurements or cost ratios are included.